A Model-Conditioned Preflight Protocol for Prior Selection in Node-Wise Scientific Latent Dynamics

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Abstract

Auxiliary priors are often introduced into scientific latent dynamics models, yet their value and interpretation are rarely clear before a full training campaign. We present a model-conditioned preflight protocol for prior selection in node-wise scientific latent dynamics. Given trajectories, a candidate graph, a fixed latent dynamics model, and a specified training budget, the protocol compares no-prior training, candidate graph Laplacian regularization, a spectrum-matched permuted graph control, optional random-graph controls, and calibrated temporal smoothing. Quick mode screens low-budget utility, standard mode tests persistence, and audit mode examines whether learned latent temporal deltas become smoother in the candidate graph basis. Across the reported regimes, spring-mass and graph-wave quick gains are explained by calibrated temporal smoothing, whereas the n-body quick run remains favorable to the candidate graph against both principal controls under the tested condition. A checkpointed harmonic-oscillator lattice audit provides a controlled positive case in which improved rollout coincides with smoother, more low-frequency latent temporal deltas on the true graph. METR-LA and graph heat show, conversely, that raw alignment or graph-generated dynamics do not guarantee candidate graph utility. The protocol is intended to separate predictive utility from topology-specific attribution and to yield an actionable prior recommendation under a fixed model condition. It is not a new graph prior, a universal physics predictor, a causal discovery method, or a state-of-the-art benchmark.

1. Introduction

Scientific dynamics models increasingly combine learned latent-state evolution with auxiliary priors motivated by domain structure. In molecular dynamics, coupled oscillator systems, particle simulations, traffic networks, and graph-discretized physical fields, observations can often be written as trajectories X_t over N interacting nodes. Before fitting a full model, a practitioner may already have a plausible candidate graph, a temporal smoothness assumption, or another regularizer that appears scientifically reasonable. Such priors are attractive because they may stabilize long-horizon rollout prediction, improve sample efficiency, and encode structure that is difficult to infer from limited data alone. Their practical value, however, is not intrinsic to the prior alone. It depends on how the prior interacts

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with the encoder, transition model, latent dimension, prior placement, optimization setting, training budget, and rollout horizon.

This dependence makes graph priors particularly difficult to interpret. A Laplacian penalty on learned node-wise latents may improve rollout because the candidate graph captures meaningful physical or scientific structure. The same gain may also arise because the penalty imposes generic graph-frequency smoothing, because a spectrum-matched permuted graph control would have produced a similar effect, because calibrated temporal smoothing would have been sufficient, or because the apparent advantage is confined to a short budget before an unconstrained model catches up. A graph-aware encoder or transition model can further blur the interpretation by already encoding part of the relational bias. Consequently, a graph prior outperforming the no-prior baseline may establish practical utility, but it does not by itself establish topology-specific attribution.

From a computational-science perspective, this ambiguity is costly. Large sweeps over priors, graph constructions, regularization strengths, seeds, and budgets consume substantial compute and can still leave attribution unresolved if the comparison set is incomplete. What is often needed first is a smaller decision stage: should the candidate graph be carried into a larger run, should a cheaper graph-free alternative be preferred, or should stronger controls be introduced before any topology-specific attribution claim is made? We therefore frame prior choice as a preflight problem. Before committing to full-scale training, a structured low-cost screen should determine whether a prior is useful, what explains the gain, and what evidence would still be required for stronger interpretation.

This article develops that idea as a model-conditioned preflight protocol for prior selection in node-wise scientific latent dynamics. The protocol compares no-prior training, a candidate graph prior, a spectrum-matched permuted graph control, optional random-graph controls, and calibrated temporal smoothing. It is organized around quick, standard, and audit modes so that predictive utility, persistence, and latent-space behavior can be examined separately. The contribution is methodological rather than architectural: the paper introduces a control-oriented workflow for prior selection and topology-specific attribution, and it uses a conservative label set to convert ambiguous prior effects into an actionable prior recommendation under a specified model condition. The manuscript does not introduce a new graph prior, claim universal physics prediction, frame the task as causal discovery, or present a state-of-the-art benchmark.

2. Related Work

This manuscript sits at the intersection of several active lines of research while remaining narrower in scope than any one of them. A first strand comes from geometric deep learning, where graph neural networks, graph-spectral constructions, and graph-based regularizers are used to encode relational inductive bias on non-Euclidean domains [1, 2]. That literature shows the value of graph structure for prediction and representation learning, but the present work does not introduce a new graph neural network or a new graph prior. Instead, it asks a different question: under a fixed latent dynamics model, when does a candidate graph prior merit further computation, and what controls are required before topology-specific attribution is justified?

A second strand concerns scientific and physical latent dynamics models, including world-model formulations that learn compact state representations and transition operators for dynamical systems [3, 4, 5]. In these settings, priors are often introduced to stabilize rollout, improve sample efficiency, or encode known structure suggested by measurement geometry or domain knowledge. The contribution here is complementary rather than competing. The paper does not propose a universal scientific world model; it proposes a standardized, low-cost preflight workflow for deciding whether a candidate prior should be carried into larger runs under a specified model condition.

A third strand includes causal and structural representation learning [6]. That literature seeks latent variables or relational structure aligned with underlying mechanisms, and it often asks whether learned representations capture more than predictive convenience. The present manuscript adopts some of the same caution about interpretation, but its goal is narrower. The workflow compares predictive priors under matched controls and asks whether observed gains support topology-specific attribution. It is not a causal discovery method, and it does not attempt to identify a true interaction graph from data alone.

A fourth strand concerns model-selection, ablation, and diagnostic workflows in machine learning and computational science. These approaches emphasize controlled baselines, matched comparisons, calibration, ablation, and concise reporting to guide computational decisions. The present protocol belongs most directly in this tradition. Its emphasis is on matched prior comparisons, persistence checks, and optional audits that produce an actionable prior recommendation before larger sweeps are undertaken, rather than on claiming state-of-the-art benchmark performance.

Taken together, these literatures motivate the need for the present study while also clarifying its boundary. This paper should be read as a prior-selection and attribution workflow for node-wise scientific latent dynamics: it draws on graph-based inductive bias, latent dynamics modeling, and diagnostic practice, but it does not claim a new graph architecture, a causal discovery result, or a state-of-the-art benchmark.

3. Problem formulation

We consider node-wise physical or scientific trajectory data $X_{1:T}$, where each frame $X_t \in \mathbb{R}^{N\times d}$ contains d observed features for each of N nodes. A dataset adapter provides these trajectories, a candidate graph or graph Laplacian $L \in \mathbb{R}^{N\times N}$, and metadata describing the graph source, the underlying simulator or dataset, and whether the graph is known, approximate, constructed, or data-derived. The predictive model maps observations into learned latent node states $H_t \in \mathbb{R}^{N\times m}$ and trains a transition model to forecast future dynamics.

The relevant object for interpretation is the model condition. In this manuscript, the model condition comprises the encoder, transition model, latent dimension, prior placement, optimizer settings, prior strength, training budget, and rollout evaluation horizon. A preflight result is therefore not a dataset-level truth statement. It is a recommendation tied to the tested model condition and must be reconsidered if the architecture, prior location, regularization strength, budget, or evaluation horizon changes.

The central question is the following:

> Under this model condition and training budget, does a candidate prior improve rollout prediction, and is the improvement better attributed to candidate topology, generic graph-frequency smoothing, calibrated temporal smoothing, or a low-budget regularization effect?

This formulation is intentionally narrower than causal discovery, universal physics modeling, or benchmark optimization. The objective is to reduce unnecessary full-training compute, avoid false topology-specific attribution, and decide whether the next step should be no prior, a candidate graph prior, calibrated temporal smoothing, or audit mode. Throughout the reported experiments, rollout quality is summarized at fixed horizons, especially $H = 16$ and $H = 32$, with $H = 32$ serving as the principal manuscript readout.

Two non-goals follow directly from this framing. First, the protocol does not claim that a useful candidate graph is physically true. Second, it does not claim universal coverage across scientific systems. The experiments are therefore presented as representative node-wise physical/scientific dynamics regimes rather than as a universal physics prediction study.

4. Mathematical and computational framing

The protocol separates prior utility from topology-specific attribution by evaluating several prior families under the same model condition, with the graph-spectral viewpoint following standard references in spectral graph theory [2]. The baseline condition is `none`, in which the latent dynamics model is trained without an auxiliary prior. The principal candidate graph prior is graph Laplacian smoothness on learned node-wise latent states. For a latent matrix H, the graph penalty is

$$R_G(H; L) = \mathrm{Tr}(H^\top L H) \tag{1}$$

If $L = U\Lambda U^\top$ is the eigendecomposition of the Laplacian and $H = UA$, then

$$\mathrm{Tr}(H^\top L H) = \mathrm{Tr}(A^\top \Lambda A) = \sum_k \lambda_k \|a_k\|_2^2 \tag{2}$$

This identity clarifies why graph-prior gains require caution. The candidate graph prior penalizes high graph frequencies more strongly, so improved rollout may reflect useful scientific structure, but it may also reflect generic smoothing in the graph spectral domain.

To probe that distinction, the key attribution baseline is the spectrum-matched permuted graph control. With permutation matrix P, the control Laplacian is

$$L_{\mathrm{perm}} = P^\top L P \tag{3}$$

This transformation preserves the eigenvalues of L, and hence the scale of graph-frequency smoothing, while disrupting node-label semantics. If the candidate graph prior improves over the no-prior baseline but not over the spectrum-matched permuted graph control, the evidence supports predictive smoothing rather than topology-specific attribution. Random-graph controls can be informative secondary checks, but they are weaker because they do not preserve the full Laplacian spectrum.

The protocol also includes a graph-free baseline,

$$R_T(H_{1:T}) = \sum_t \|H_{t+1} - H_t\|_F^2 \tag{4}$$

which we refer to as calibrated temporal smoothing when its effective regularization scale has been matched to the candidate graph prior. Calibration is necessary because raw temporal losses can be much smaller than graph-Laplacian losses. If α_G denotes the graph-prior weight and R_G^0 and R_T^0 denote initial auxiliary-loss values measured under the same calibration pass, the temporal weight is chosen as

$$\alpha_T = \frac{\alpha_G R_G^0}{R_T^0 + \varepsilon} \tag{5}$$

so that

$$\alpha_T R_T^0 \approx \alpha_G R_G^0 \tag{6}$$

This calibration does not assert that graph and temporal priors are equivalent. It ensures that the comparison asks a fairer question: at matched initial regularization strength, does the candidate graph add value beyond graph-free temporal smoothing? The same framing also clarifies why graph versus none alone is insufficient for topology-specific attribution. That stronger interpretation requires explicit comparison with the spectrum-matched permuted graph control and with calibrated temporal smoothing.

5. Model-conditioned preflight protocol

The model-conditioned preflight protocol is designed as a staged computational screen rather than a substitute for full training. It begins with diagnostics that summarize the candidate graph and the observed dynamics, proceeds to controlled mini-training runs, and escalates to longer budgets or latent audits only when the quick result warrants additional analysis.

Operationally, the workflow has four stages. First, trajectories, candidate graph metadata, and the corresponding Laplacian are loaded from a dataset adapter. Second, the prior family set is defined: `none`, the candidate graph prior, the spectrum-matched permuted graph control, optional random-graph controls, and calibrated temporal smoothing when a graph-versus-temporal comparison is needed. Third, models are trained under quick mode and, when justified, under standard mode; rollout is evaluated at fixed horizons. Fourth, if checkpointed latent traces are available, audit mode examines whether rollout gains coincide with smoother or more low-frequency latent temporal deltas in the candidate graph basis.

Figure 1 summarizes this workflow.

The distinction between operating modes is crucial. Quick mode asks whether any prior is useful as a low-cost regularizer. Standard mode asks whether the quick result persists once the base model has more time to fit the dynamics. Audit mode asks whether predictive gains coincide with latent behavior aligned with the candidate graph. Figure 2 shows how these comparisons lead to topology-specific attribution only when the relevant controls have been cleared.

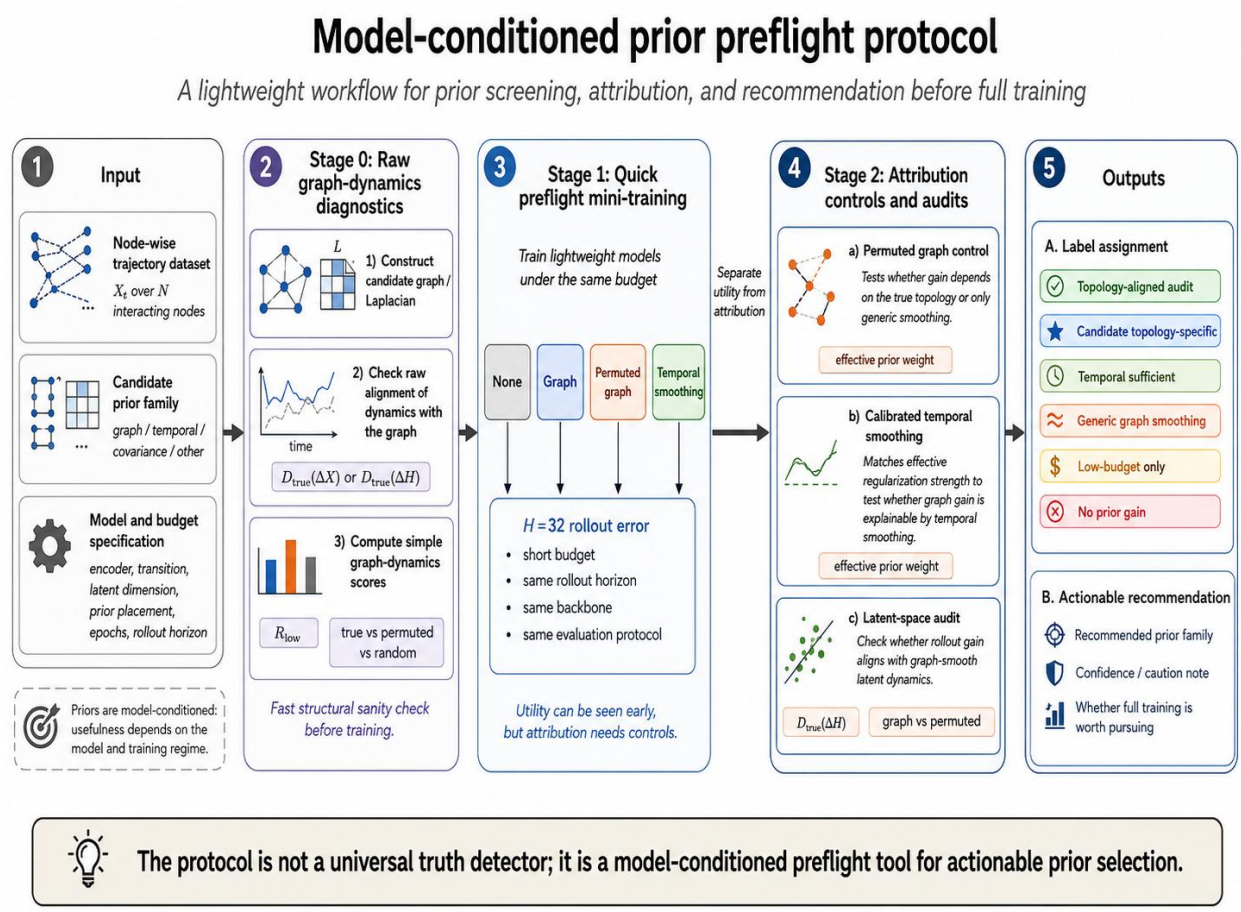

Figure 1: Model-conditioned preflight workflow. The protocol converts prior selection into a staged computational decision. Node-wise trajectories and a candidate graph first pass through raw diagnostics, then quick preflight comparisons, then spectrum-matched and calibrated controls. Standard and audit modes are used only when the quick result warrants persistence or mechanism checks. The output is an actionable prior recommendation rather than a claim that the candidate graph is physically true.

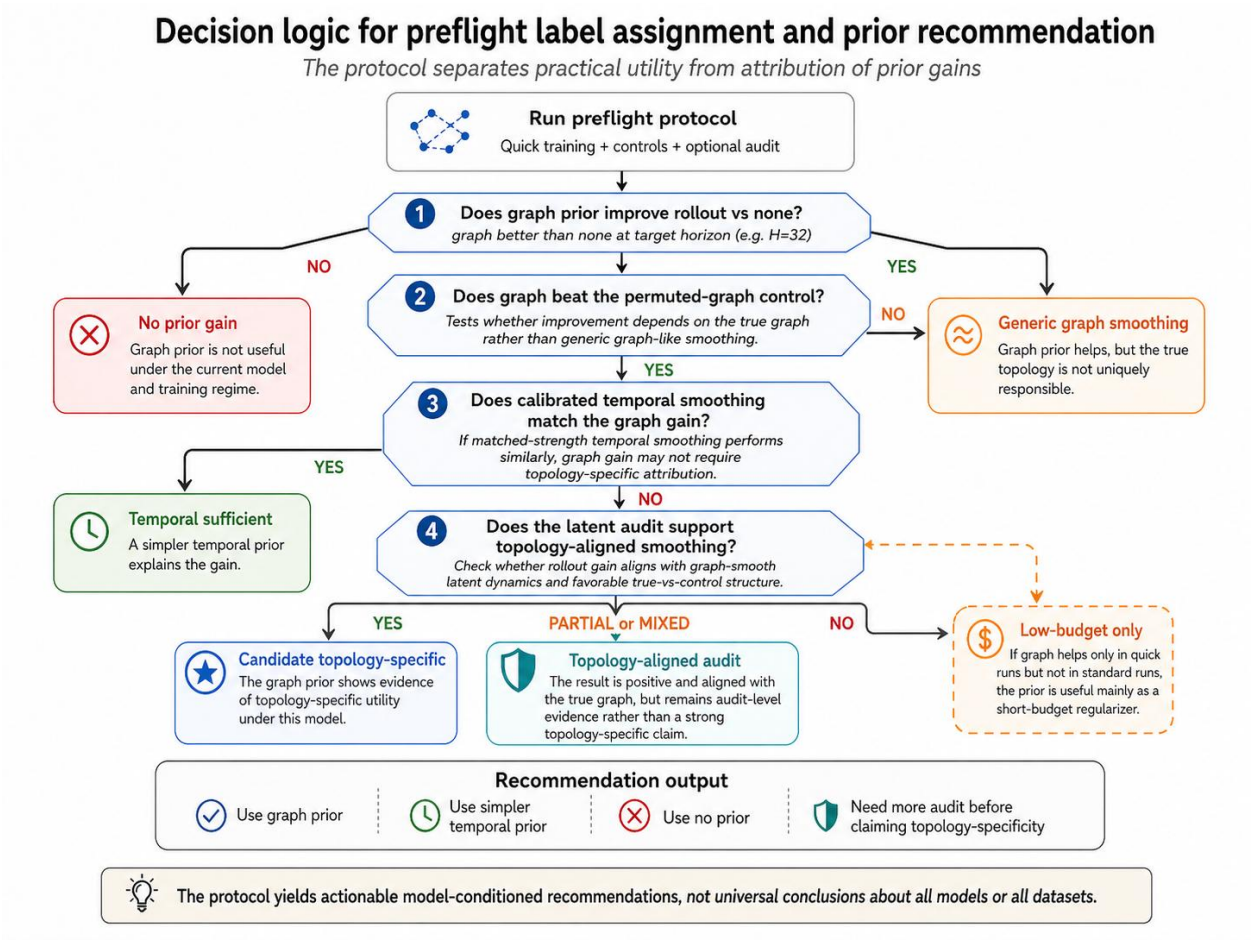

Figure 2: Decision logic for label assignment and prior recommendation. The decision tree separates predictive utility from topology-specific attribution. A candidate graph must first improve over no-prior training, then beat the spectrum-matched permuted graph control and calibrated temporal smoothing before it supports topology-specific attribution under the tested condition; audit mode provides the stronger evidence of topology-aligned latent smoothing. A graph outperforming the no-prior baseline alone is therefore not sufficient evidence of topology-specific attribution.

The resulting labels are listed in Table 1, and the prior families on which they depend are summarized in Table 2. Taken together, these tables define the vocabulary of the protocol and the corresponding actionable prior recommendation associated with each outcome.

Table 1: Protocol labels and corresponding actionable prior recommendations

display label	**evidence pattern**	**recommended action**	**claim allowed**	**claim boundary**
no prior gain	No tested prior improves meaningfully over the no-prior model.	Skip the prior for larger runs unless there is a separate scientific reason to test it.	The tested prior was not useful under the model condition.	Does not imply the prior is useless for all models or budgets.
generic smoothing	The candidate graph improves over no-prior training but does not beat the spectrum-matched permuted graph control or comparable smoothing controls.	Treat the result as smoothing-like and test cheaper or simpler controls first.	Graph-style regularization helped prediction.	Do not claim topology-specific attribution.
temporal sufficient	Calibrated temporal smoothing matches or beats the candidate graph prior.	Prefer calibrated temporal smoothing unless topology-specific attribution is itself the objective.	Graph-free temporal smoothing explains the observed gain.	Do not claim that graph structure is necessary.
candidate topology	The candidate graph beats no-prior training, the spectrum-matched permuted graph control, and calibrated temporal smoothing under the tested budget.	Carry the candidate graph forward with more seeds, stronger controls, and audit where possible.	The result supports topology-specific attribution under the tested condition.	It does not prove that the candidate graph is the true physical interaction graph.
topology-aligned audit	Rollout gains coincide with smoother or more low-frequency learned latent temporal deltas in the candidate graph basis.	Use as the strongest preflight support for continuing the candidate graph.	Audit evidence supports topology-aligned latent smoothing under the tested condition.	The result remains model-conditioned, weight-conditioned, and budget-conditioned.
low-budget only	A prior helps in quick mode but loses the advantage under standard or longer training.	Treat the prior as a low-budget regularizer rather than a persistent structural advantage.	The prior can help under constrained training.	Do not extrapolate the effect to longer budgets.
overconstrained	A prior harms rollout relative to no-prior training and controls.	Reduce strength, revise prior placement, or skip the prior.	The tested prior suppressed useful predictive variation.	This does not rule out weaker or differently placed priors.
inconclusive	Required controls, seeds, or audit artifacts are insufficient.	Run the missing controls or avoid stronger claims.	Evidence is insufficient for a stronger label.	Do not over-interpret incomplete comparisons.

Table 2: Prior families and attribution roles

prior family	**definition**	**attribution role**	**manuscript use**
`none`	No auxiliary prior.	Establishes whether any prior improves rollout.	Baseline for all preflight recommendations.

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prior family	definition	attribution role	manuscript use
`graph_laplacian`	Candidate-graph Laplacian smoothness on learned node-wise latents, $\mathrm{Tr}\left(H^\top L H\right)$.	Tests whether candidate graph-frequency smoothing helps the tested model.	Main prior under evaluation, not introduced as a new prior.
`permuted_graph`	Spectrum-matched permuted Laplacian, $L_{\mathrm{perm}} = P^\top L P$.	Preserves Laplacian spectrum while removing node-label semantics.	Decisive control for topology-specific attribution.
`random_graph`	Optional graph with matched coarse size or edge statistics.	Tests whether an unrelated graph can produce similar smoothing.	Secondary control; weaker than the spectrum-matched permuted graph control.
`temporal_smooth`	Graph-free smoothness on learned latent temporal deltas, e.g. $\sum_t \lVert H_{t+1} - H_t \rVert_F^2$.	Tests whether temporal regularization explains candidate graph gains.	Must be calibrated when compared with graph priors.
future priors	Planned auxiliary baselines for additional scientific structure.	Broadens the preflight family set.	Mentioned as future extensions, not current core evidence.

Implementation Overview

In practice, the preflight workflow is executed through an adapter-based preflight script. Each adapter is responsible for dataset-specific trajectory loading and for supplying the candidate graph information needed by the model-conditioned comparison. Depending on the regime, this means either loading a candidate graph Laplacian directly or constructing the candidate graph Laplacian from dataset-specific metadata before the matched prior comparisons are run.

This design keeps the comparison logic uniform across datasets. Once trajectories, the candidate graph Laplacian, and the associated metadata have been exposed through the adapter interface, the same preflight routine can evaluate no-prior training, the candidate graph prior, the spectrum-matched permuted graph control, optional random-graph controls, and calibrated temporal smoothing under a shared model condition. The resulting artifacts are written in a standardized form: `run_config.json` records the run specification, `summary.csv` provides the compact metric summary, and `preflight_report.md` provides the human-readable interpretation and actionable prior recommendation.

The implementation is also intended to make extension straightforward without changing the underlying methodology. New dataset support is added through adapters rather than by rewriting the core preflight procedure, so the same low-cost screening can be applied before larger sweeps, longer budgets, or expanded seed studies. In that sense, the implementation overview is part of the methodological contribution: it operationalizes standardized low-cost screening while keeping dataset-specific loading and candidate graph construction isolated from the matched prior comparisons themselves.

6. Experimental regimes and implementation

The experiments are presented as a stress test of the preflight methodology rather than as a leaderboard. All reported values are drawn from existing completed runs and sum-

maries documented in the accompanying source materials; no new experiments are introduced in this revision. The purpose of the experimental section is to ask whether a single model-conditioned protocol can return sensible decisions across representative node-wise physical/scientific dynamics regimes.

The benchmark set spans controlled oscillator lattices, spring-mass systems, graph-wave and graph heat lattices, a graph low-frequency stress test, approximate n-body interactions, molecular dynamics summaries related to MD17/rMD17 [7, 8], and a traffic forecasting case connected to METR-LA as popularized by DCRNN [9]. These regimes differ not only in the underlying dynamics but also in candidate graph provenance: some graphs are simulator-defined, some are approximate physical constructions, some are synthetic stress tests, and some are data-derived. This heterogeneity is intentional because the provenance of the candidate graph is part of the interpretation boundary.

Implementation settings are fixed within each model condition. Quick mode corresponds to the short training budget used for screening, typically five epochs in the reported cases; standard mode uses the longer budget available for persistence checks, typically twenty epochs where such runs were completed; and audit mode is used only when checkpointed latent traces are available. Rollout comparisons focus on $H = 32$, which is the central summary statistic used throughout the figures and tables. Table 3 summarizes the regimes and the role each plays in the manuscript, while detailed file-level traceability is retained in Appendix D.

Table 3: Benchmark regimes used for preflight stress testing

regime	**graph source**	**role in manuscript**	**key preflight readout**
HO lattice	true lattice	Controlled positive audit case.	The candidate graph improves H32 rollout and the Cycle 8 audit shows smoother, more low-frequency latent deltas on the true graph.
Spring-mass lattice	true spring lattice	Second-order mechanics and temporal-baseline check.	The quick candidate graph gain is explained by calibrated temporal smoothing; the standard budget favors no-prior training.
Graph-wave lattice	true wave lattice	PDE-like propagation and temporal-baseline check.	The quick candidate graph gain is smoothness-like; calibrated temporal smoothing slightly beats the candidate graph; the standard budget favors no-prior training.
N-body distance	initial distance-kNN weighted graph	Approximate long-range interaction graph.	The quick candidate graph beats no-prior training, the spectrum-matched permuted graph control, and calibrated temporal smoothing; the standard budget favors the spectrum-matched permuted graph control.

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regime	graph source	role in manuscript	key preflight readout
METR-LA correlation graph	data-derived `correlation_topk` sensor graph	Real traffic application case.	Raw alignment does not translate into rollout gain; the candidate graph is worse than no-prior training and the spectrum-matched permuted graph control.
Graph heat lattice	true generator graph	Cautionary graph-generated case.	The true generator graph does not help the tested latent model; the candidate graph harms H32 rollout.
Graph low-frequency lattice	constructed low-frequency Laplacian basis	Synthetic alignment stress test.	Strong raw alignment does not guarantee quick topology-specific attribution; budget dependence appears in follow-up analysis.
rMD17 molecular dynamics summaries	molecular cutoff graphs	Earlier graph-style smoothing and specificity stress test.	Graph-style priors improve rollout, but exact molecular topology-specific attribution is weak against controls.

7. Results

Across the reported regimes, the model-conditioned protocol yields different actionable prior recommendations rather than a single preferred prior family. Figure 3 makes this point at a glance: spring-mass and graph-wave favor calibrated temporal smoothing in quick mode, the n-body quick run supports topology-specific attribution under the tested condition, the HO lattice provides the controlled positive audit case, and METR-LA together with graph heat recommend against carrying the candidate graph forward. The interpretation is that the protocol distinguishes predictive utility from topology-specific attribution across regimes rather than merely ranking priors. The boundary is that detailed case-level evidence for the graph low-frequency lattice and the rMD17 summaries is retained in Appendix D rather than in the main text.

A first substantive result is that calibrated temporal smoothing changes the interpretation of the spring-mass and graph-wave quick runs. In spring-mass, the candidate graph improves $H = 32$ rollout from 0.6864 with no-prior training to 0.5740, but calibrated temporal smoothing reaches 0.5689; in graph-wave, the candidate graph improves $H = 32$ from 0.6825 to 0.5712, but calibrated temporal smoothing reaches 0.5663. The interpretation is that the observed low-budget gains are real but do not require topology-specific attribution, because a graph-free baseline explains them once prior strength is calibrated. The boundary is explicit: these cases support `temporal_smoothing_sufficient`, not a claim that the candidate graph carries indispensable structural information under the tested condition.

A second result is that the n-body quick run provides the strongest main-text case for topology-specific attribution under a quick budget. At $H = 32$, no-prior training yields 0.8216, the candidate graph 0.5433, the spectrum-matched permuted graph control 0.5838, and calibrated temporal smoothing 0.6453. The interpretation is that, under this model-conditioned quick run, the candidate graph remains favorable even after the principal attribution controls are applied, so carrying it forward is justified. The boundary is equally important: this result does not show that the distance-kNN graph is the true physical interaction graph, and it does not by itself establish persistence at longer budgets.

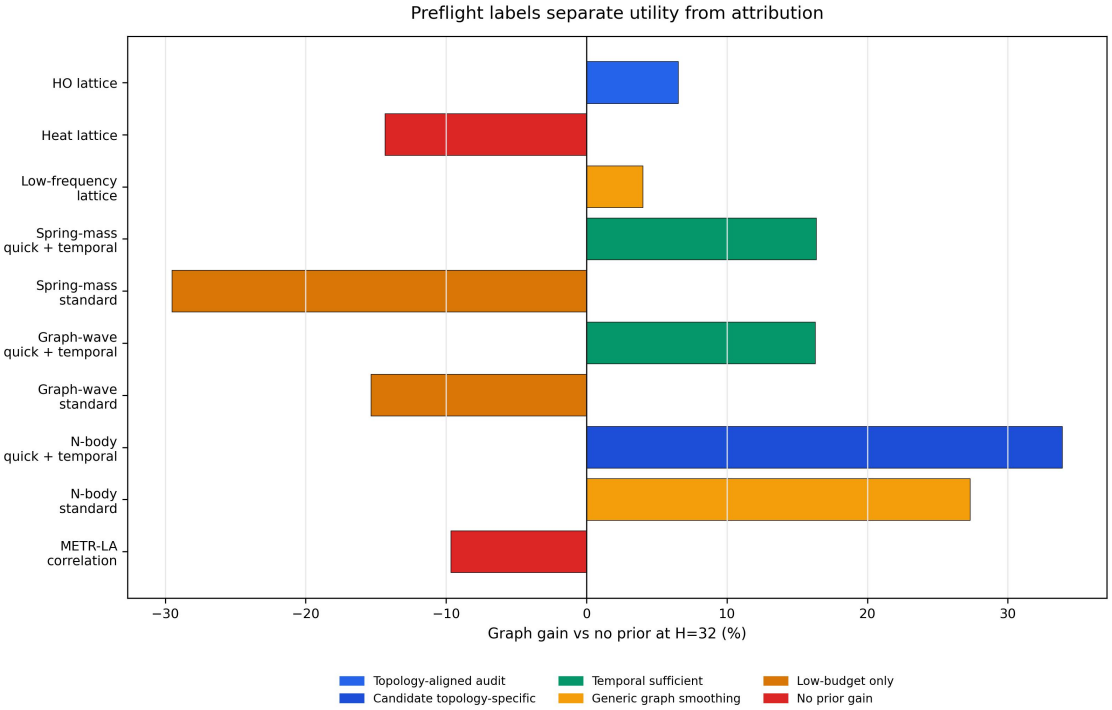

Figure 3: Preflight classification overview across representative regimes. The same model-conditioned protocol returns different computational recommendations across representative node-wise physical/scientific dynamics regimes: skip the prior when the candidate graph harms rollout, prefer calibrated temporal smoothing when it explains the gain, continue the candidate graph when it beats the principal controls, and reserve the strongest claim for the audited positive case. The figure emphasizes that the contribution is a decision workflow, not a universal ranking of priors.

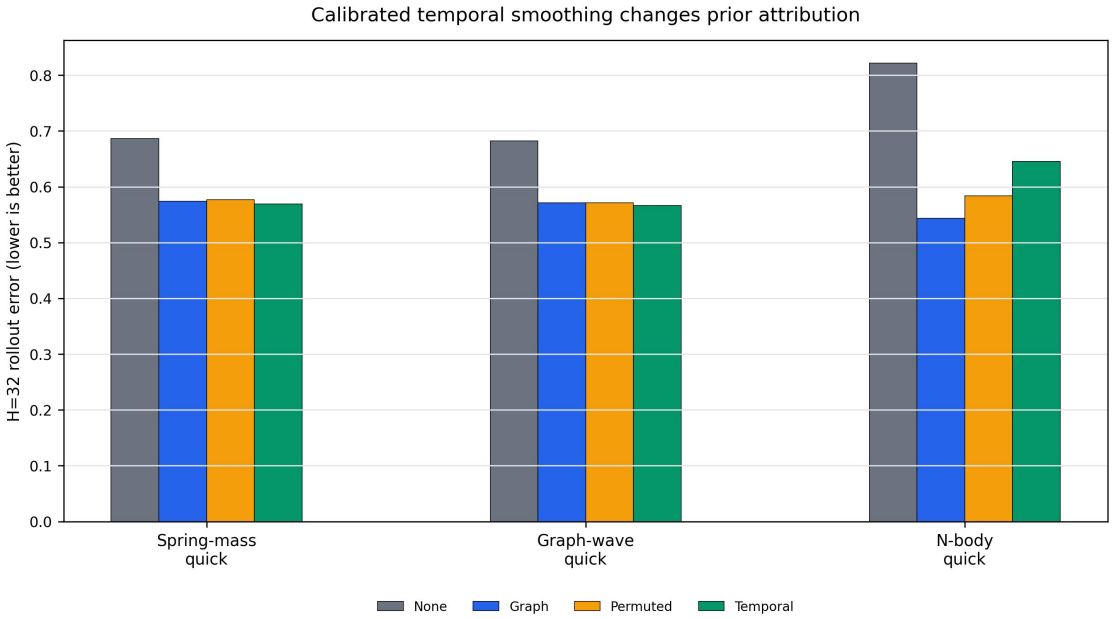

Figure 4: Calibrated prior-family comparison. Spring-mass and graph-wave show that calibrated temporal smoothing can explain apparent candidate graph gains, whereas the n-body quick run remains favorable to the candidate graph against both the spectrum-matched permuted graph control and calibrated temporal smoothing. The figure therefore distinguishes predictive utility from topology-specific attribution.

A third result is that budget must be treated as part of the model condition. In spring-mass, the quick candidate graph gain does not persist: at the standard budget, $H = 32$ is 0.0809 for no-prior training and 0.1048 for the candidate graph. Graph-wave behaves similarly, with 0.0826 for no-prior training and 0.0953 for the candidate graph. In n-body, the standard-budget run favors the spectrum-matched permuted graph control at 0.0890, compared with 0.1135 for the candidate graph. The interpretation is that some quick-mode positives are low-budget regularization effects rather than persistent structural advantages. The boundary is that quick-mode evidence is still useful for screening, but it should not be treated as budget-robust support for topology-specific attribution.

A fourth result is that audit mode provides the strongest positive evidence when rollout and latent-space diagnostics agree. In the checkpointed HO lattice audit, the candidate graph achieves $H = 32$ rollout error 0.0169, compared with 0.0198 for the spectrum-matched permuted graph control and 0.0199 for the random-graph control; the normalized $D_true($ $DeltaH)$ is 2.8106 for the candidate graph and 3.5109 for the spectrum-matched permuted graph control, and R_{low} at $K = 4$ is 0.1866 versus 0.1216. The interpretation is that improved rollout coincides with smoother and more low-frequency latent temporal deltas in the true-graph basis, which is precisely the pattern targeted by `topology_aligned_latent_smoothing`. The boundary is that this remains a controlled positive case under the tested condition rather than a universal latent-mechanism claim.

A fifth result is that the negative cases are integral to the method rather than peripheral to it. In METR-LA, the candidate graph does not improve predictive performance: $H = 32$ is 0.4068 for no-prior training, 0.4461 for the candidate graph, and 0.4337 for the spectrum-matched permuted graph control. In graph heat, the candidate graph again fails, with

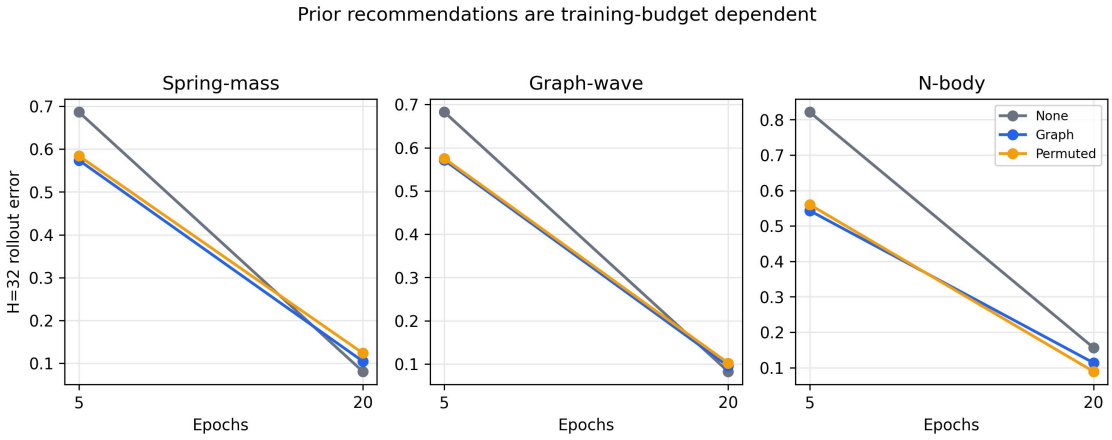

Figure 5: Training-budget dependence. Quick candidate graph gains do not necessarily persist under standard training. Spring-mass and graph-wave favor no-prior training at twenty epochs, and n-body favors the spectrum-matched permuted graph control at the longer budget. The figure illustrates why the protocol treats budget as part of the model condition.

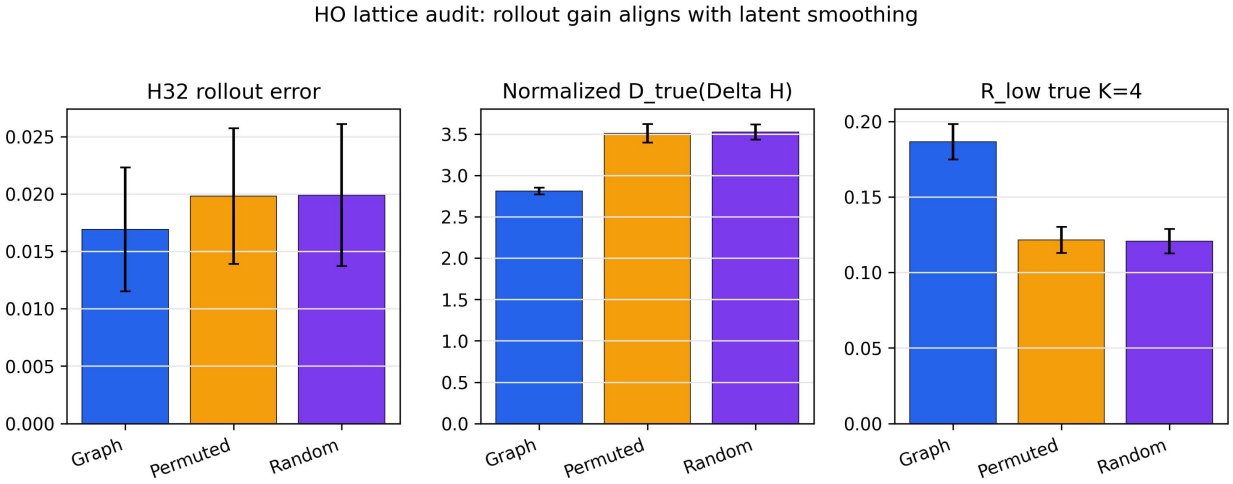

Figure 6: Latent audit positive case. The checkpointed HO lattice audit shows the strongest form of support for a candidate graph: improved rollout coincides with smoother and more low-frequency latent temporal deltas in the true-graph basis. This is presented as a controlled positive case for audit mode rather than as a universal mechanism claim.

$H = 32$ equal to 0.6036 for no-prior training, 0.6902 for the candidate graph, and 0.6058 for the spectrum-matched permuted graph control. The interpretation is that raw alignment or graph-generated dynamics do not guarantee candidate graph utility in the tested latent model. The boundary is that these results are model-conditioned cautionary cases rather than general prohibitions on graph priors.

8. Discussion

The main contribution of the manuscript is methodological. It provides a principled way to decide, before expensive training, whether a candidate graph should be pursued, whether calibrated temporal smoothing is sufficient, or whether stronger controls are required before any topology-specific attribution is made. In that sense, the protocol addresses a recurring problem in computational science: prior choice is consequential, but prior interpretation is easy to overstate.

The reported cases sharpen the distinction between predictive utility and topology-specific attribution. A candidate graph beating the no-prior baseline can be useful, but it has not yet shown that node-label structure matters. That stronger interpretation requires the spectrum-matched permuted graph control and calibrated temporal smoothing, and it becomes more compelling when audit mode shows that learned latent temporal deltas align with the candidate graph. The n-body quick run and the HO audit illustrate these two levels of evidence, while spring-mass and graph-wave show how the same workflow can recommend a cheaper graph-free alternative.

The budget comparison further clarifies the role of preflight. Quick mode is a screen for whether a prior deserves additional computational attention, not a substitute for persistence checks. The negative cases reinforce the same point from the opposite direction: METR-LA and graph heat show that raw graph alignment or graph-generated dynamics are not sufficient reasons to keep a candidate graph. The value of the protocol, therefore, lies as much in ruling out unsupported interpretations as in identifying promising priors.

Table 4 summarizes the principal claims, their evidentiary basis, and the boundaries that accompany them. Detailed case-level traceability is retained in Appendix D.

Table 4: Claims, evidence, and boundaries

manuscript claim	**supporting evidence**	**key numbers**	**boundary**
The preflight protocol gives actionable prior recommendations.	Preflight benchmark summary and individual preflight reports.	Labels include no gain, temporal sufficient, candidate topology-specific, generic smoothing, and audit-supported positive cases.	Recommendations apply only under tested model and budget conditions.
The spectrum-matched permuted graph control is necessary for topology-specific attribution.	Master decision rules and graph specificity table.	HO lattice lambda=0.1 beats the spectrum-matched permuted graph control at H32; many rMD17 cases do not.	A candidate graph beating no-prior training alone is not topology-specific attribution.
Calibrated temporal smoothing changes candidate graph attribution.	Calibrated spring-mass, graph-wave, and n-body preflight reports.	Spring-mass temporal H32 0.5689 vs graph 0.5740; graph-wave temporal H32 0.5663 vs graph 0.5712; n-body graph H32 0.5433 vs temporal 0.6453.	Calibration is required because raw prior losses differ by scale.

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manuscript claim	supporting evidence	key numbers	boundary
Candidate-graph gains can be low-budget-only.	Quick/standard comparisons for spring-mass, graph-wave, and n-body.	Spring-mass graph H32 quick 0.5740 vs standard 0.1048, while standard no-prior is 0.0809; graph-wave standard no-prior is best; n-body standard spectrum-matched permuted graph control is best.	This does not imply that graph priors never help longer runs.
The n-body quick run supports topology-specific attribution under the tested condition.	N-body quick calibrated preflight report.	H32 none 0.8216, graph 0.5433, permuted control 0.5838, temporal 0.6453.	This is not proof that the distance-kNN graph is the physical interaction graph.
Audit mode can support topology-aligned latent smoothing.	Cycle 8 checkpointed HO lattice audit.	H32 graph 0.0169, permuted control 0.0198, random 0.0199; graph $D_{\text{true,norm}}(\Delta H)$ 2.8106 vs permuted control 3.5109; graph R_{low} at $K = 4$ 0.1866 vs 0.1216.	This is a controlled positive case, not universal latent-alignment evidence.
Raw graph-dynamics alignment is not sufficient.	Cycle 6 diagnostics, METR-LA, graph low-frequency lattice, graph heat.	METR raw true $D_{dX,\text{norm}}$ 0.2380 vs permuted 0.3044, but graph H32 is worse than no-prior training; the graph low-frequency case also fails to establish quick topology-specific attribution.	Raw diagnostics are supporting probes, not decision rules.
Graph-generated dynamics do not guarantee candidate graph utility.	Graph heat lattice report and failure audit.	H32 none 0.6036, graph 0.6902, permuted control 0.6058.	The prior acts on learned latents, not directly on the simulator equation.

9. Limitations

The protocol is intentionally conservative. Its recommendations are model-conditioned and budget-conditioned: they apply to the tested encoder, transition model, latent dimension, prior placement, optimizer settings, prior strength, training budget, and rollout horizon. A recommendation should therefore be treated as a local computational conclusion rather than as a property of the dataset alone.

Several methodological boundaries follow from this framing. Candidate graphs in the reported regimes range from simulator-defined lattices to approximate physical constructions and data-derived correlation graphs, and those categories should not be interpreted in the same way. The spectrum-matched permuted graph control is the strongest control used here, but it does not exhaust every possible graph confound. Calibrated temporal smoothing is required for fair graph-versus-temporal comparison. Raw-coordinate diagnostics such as $D_L($ $DeltaX)$ can be useful supporting probes, but because the prior acts on learned latents, they are not sufficient decision rules.

A further limitation is that audit mode depends on checkpointed latent traces or saved latent artifacts. Where those artifacts are missing, mechanism claims must remain limited. Finally, the benchmark spans representative node-wise physical/scientific dynamics regimes rather than the whole space of scientific prediction problems. The manuscript should therefore not be read as introducing a universal physics predictor, a causal discovery method, or a state-of-the-art benchmark.

Table 5 consolidates these boundaries in the form used throughout the draft.

Table 5: Limitations and claim boundaries

limitation or boundary	**boundary statement**	**implication**
Model conditioning	Recommendations apply under the tested encoder, transition model, prior placement, prior strength, optimizer, training budget, and rollout horizon.	Changing the model condition requires a new preflight interpretation.
Budget conditioning	Quick-mode gains can disappear under standard or longer training.	Quick positives should be validated before stronger claims.
Candidate graph ambiguity	Candidate graphs may be true, approximate, constructed, or data-derived.	A useful candidate graph prior does not prove physical truth.
Control requirements	Topology-specific attribution requires beating the spectrum-matched permuted graph control and calibrated temporal smoothing.	Graph versus none is insufficient for topology-specific attribution.
Temporal calibration	Raw temporal prior losses may be much smaller than graph-prior losses.	Temporal baselines should be calibrated before attribution.
Raw diagnostics	Raw-coordinate smoothness can disagree with learned latent prior utility.	Raw $D_L(\Delta X)$ should not be used as a sufficient decision rule.
Audit artifacts	Latent audits require saved traces or checkpoints.	Missing artifacts limit mechanism claims.
Benchmark scope	Regimes are representative stress tests, not universal physics coverage.	Avoid universal predictor or SOTA benchmark framing.
Causal claims	The workflow compares predictive priors and controls.	Do not frame the result as causal discovery.
Prior novelty	The graph Laplacian prior is a tested prior family, not the paper's algorithmic novelty.	The contribution is the preflight protocol and attribution workflow.

10. Conclusion

This manuscript presents a model-conditioned preflight protocol for prior selection in node-wise scientific latent dynamics. By combining no-prior baselines, a candidate graph prior, the spectrum-matched permuted graph control, calibrated temporal smoothing, and audit mode when latent artifacts are available, the protocol turns prior choice into a structured computational decision rather than an ad hoc modeling preference.

Across the reported regimes, the pattern is consistent. Candidate graphs can help under low training budgets, but calibrated temporal smoothing and the spectrum-matched permuted graph control often explain the gain, and longer training can change the recommendation. Under the tested quick-budget condition, the n-body case remains the clearest example supporting topology-specific attribution, while the HO lattice audit provides the strongest positive example of topology-aligned latent smoothing. Spring-mass and graph-wave show when calibrated temporal smoothing is sufficient, and METR-LA together with graph heat show when the appropriate actionable prior recommendation is to avoid the candidate graph. The contribution is therefore a practical workflow for prior selection and interpretation, not a new graph prior, a universal physics predictor, a causal discovery method, or a state-of-the-art benchmark.

Data and Code Availability

This manuscript reports previously completed runs only and introduces no new experiments in this version. The preflight tool is implemented in the project repository as an adapter-based script in which dataset-specific adapters provide trajectories and candidate graph information, while the shared routine writes `run_config.json`, `summary.csv`, and `preflight_report.md` as standardized run artifacts. Code and manuscript assets are available at https://github.com/2GyounnnG/causalworld. A versioned archival snapshot will be provided upon publication or repository release.

Appendix A. Protocol details and operating modes

The protocol is best understood as a sequence of increasingly demanding questions. Quick mode asks whether any prior helps under a short budget. Standard mode asks whether that conclusion persists when training is allowed to continue. Audit mode asks whether predictive gains coincide with latent behavior that is more consistent with the candidate graph. The strongest label therefore requires agreement across utility, control comparisons, and latent-space evidence.

The labels in Table 1 should be interpreted as report-level decisions rather than as permanent properties of a dataset. They are intentionally conservative: `no_prior_gain` and `overconstrained` recommend against continuing the prior; `temporal_smoothing_sufficient` recommends a cheaper graph-free alternative; `candidate_topology_specific` supports topology-specific attribution under the tested condition; and `topology_aligned_latent_smoothing` requires audit evidence in addition to predictive gains.

Appendix B. Benchmark regimes and dataset-adapter notes

The benchmark set mixes graphs with different provenance. HO, spring-mass, graph-wave, and graph heat use true or simulator-defined lattices. The graph low-frequency regime uses a constructed spectral stress test. N-body uses an approximate distance-kNN graph. The rMD17 summaries use molecular cutoff graphs. METR-LA uses a correlation-derived sensor graph. This variation is intentional because candidate graph provenance is part of the interpretation boundary addressed by the preflight protocol.

Appendix C. Prior families and calibration formulas

The graph Laplacian prior on learned node-wise latent states is

$$R_G(H; L) = \mathrm{Tr}(H^\top L H) \tag{C.1}$$

The spectrum-matched permuted graph control is

$$L_{\mathrm{perm}} = P^T L P, \tag{C.2}$$

where P is a permutation matrix. This preserves Laplacian eigenvalues while disrupting node-label semantics.

The graph-free temporal smoothness prior is

$$R_T(H_{1:T}) = \sum_t \|H_{t+1} - H_t\|_F^2 \tag{C.3}$$

For calibrated prior-strength matching, let α_G be the graph-prior weight, and let R_G^0 and R_T^0 denote initial graph and temporal prior losses under the same calibration pass. The calibrated temporal weight is

$$\alpha_T = \frac{\alpha_G R_G^0}{R_T^0 + \varepsilon} \tag{C.4}$$

so that

$$\alpha_T R_T^0 \approx \alpha_G R_G^0 \tag{C.5}$$

Appendix D. Claim traceability and supporting evidence

This appendix provides a compact traceability table linking each major manuscript claim to the supporting experiment summaries, key numerical evidence, and interpretation caveats.

Table D.6: Claim traceability and supporting evidence

claim	**supporting evidence**	**key numbers**	**caveats**	**appears in**
The paper contributes a computational preflight protocol for model-conditioned prior selection.	`paper/jcs_outline.md`; `paper/jcs_draft_intro.md`; `paper/jcs_draft_methods.md`	Model condition includes encoder, transition model, prior placement, strength, training budget, and rollout horizon.	Not a new graph prior or SOTA benchmark.	Figure 1; Tables 1-2
The spectrum-matched permuted graph control is the key topology-specific attribution control.	`paper/jcs_draft_methods.md`; `paper/tables/evidence_map.md`	$L_{\text{perm}} = P^\top LP$; HO lattice lambda=0.1 permuted-control delta 0.002866 at H32 with CI $[0.001070, 0.004210]$.	Random graph controls are secondary and not fully spectrum-matched.	Figure 2; Tables 2 and 4
Calibrated temporal smoothing can explain some candidate graph gains.	`paper/tables/evidence_map.md`; `paper/tables/table4_claims_evidence.md`	Spring-mass temporal H32 0.5689 vs graph 0.5740; graph-wave temporal H32 0.5663 vs graph 0.5712.	Requires calibrated prior strength; uncalibrated temporal comparisons are not enough.	Figure 4; Tables 1 and 4
The n-body quick run supports topology-specific attribution under the tested condition.	`paper/tables/evidence_map.md`; `paper/tables/table4_claims_evidence.md`	H32 none 0.8216, graph 0.5433, permuted 0.5838, temporal 0.6453.	Candidate graph is distance-kNN and approximate; not proof of true physical graph.	Figures 3-4; Tables 3-4
Candidate-graph benefits can be low-budget-only.	`paper/tables/evidence_map.md`; `paper/tables/table4_claims_evidence.md`	Spring-mass standard no-prior H32 0.0809 vs graph 0.1048; graph-wave standard no-prior 0.0826 vs graph 0.0953; n-body standard permuted 0.0890 vs graph 0.1135.	Does not rule out graph priors under other models or longer audits.	Figure 5; Tables 1 and 4

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claim	supporting evidence	key numbers	caveats	appears in
HO lattice/Cycle 8 is a controlled positive latent-audit case.	`paper/tables/evidence_map.md`; `paper/figure_data/fig6_latent_audit_positive_case.csv`	H32 graph 0.0169, permuted 0.0198, random 0.0199; $D_{\mathrm{true,norm}}(\Delta H)$ graph 2.8106, permuted 3.5109, random 3.5237; R_{low} true $K = 4$ graph 0.1866, permuted 0.1216.	High-prior-weight controlled case; not universal latent mechanism evidence.	Figure 6; Tables 3-4
METR-LA correlation graph demonstrates no graph gain despite raw alignment.	`paper/tables/evidence_map.md`; `paper/tables/table4_claims_evidence.md`	Raw true $D_{dX,\mathrm{norm}}$ 0.2380, permuted 0.3044, random 0.4737; H32 none 0.4068, graph 0.4461, permuted 0.4337.	Uses `correlation_topk` graph, not official road adjacency.	Figure 3; Tables 3-5
Graph heat lattice shows true generator graph does not guarantee prior utility.	`paper/tables/evidence_map.md`; `paper/tables/table4_claims_evidence.md`	H32 none 0.6036, graph 0.6902, permuted 0.6058.	Applies to the tested latent model and prior setting.	Figure 3; Tables 3-5
rMD17 supports graph-style smoothing but weak exact molecular graph specificity.	`paper/tables/evidence_map.md`; `paper/tables/table4_claims_evidence.md`	Cycle 2 H32 graph gains versus none: aspirin +67.2%, ethanol +19.4%, malonaldehyde +49.6%, naphthalene +67.0%, toluene +47.0%; controls often match or beat graph.	Earlier cycles are attribution context, not the core preflight benchmark table.	Tables 3-4 and discussion

Appendix E. Reproducibility note and software interface

The present draft summarizes previously completed runs only. The software interface follows an adapter-based design in which dataset-specific trajectory loading and candidate graph Laplacian construction or loading are handled by adapters, while the core routine carries out the matched prior comparisons under a shared model condition. In this interface, `run_config.json` records the run specification, `summary.csv` provides the compact metric summary, and `preflight_report.md` provides the human-readable report and actionable prior recommendation. Companion documentation targets for this tool include `paper/tool_readme_draft.md` and `docs/graph_prior_preflight_tool.md`, alongside the command-line example below.

```
conda run -n causalworld python scripts/graph_prior_preflight_check.py \
  --dataset spring_mass_lattice \
  --prior-weight 0.1 \
  --epochs 5 \
  --train-transitions 96 \
  --eval-transitions 32 \
  --horizons 16 32 \
  --include-temporal-prior \
  --calibrate-prior-strength \
  --out-dir analysis_out/preflight_runs/spring_mass_lattice_ep5_temporal_calibrated
```

This command is included for interface clarity only; no new experiment is introduced in this manuscript version.

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